Arista Access Node WFD-ACC-59 Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) SOP

Node: WFD-ACC-59 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_2307

Description

This Standard Operating Procedure (SOP) addresses the occurrence of a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain on the Arista Access Node WFD-ACC-59 at the Whitefield POP site. The failure is attributed to the inherent design and operational complexities of the Access Node's role in the network topology. Specifically, the Arista Access Node, being a critical component of the network's edge infrastructure, is responsible for routing traffic between the core network and the access network. The occurrence of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on this node can be attributed to various factors, including but not limited to, misconfigured routing policies, suboptimal network design, or hardware-related issues. The BGP route churn can be caused by changes in the network topology, such as the addition or removal of prefixes, which can lead to flapping of routes. OSPF adjacency loss can be caused by issues with the OSPF protocol configuration or network instability. Meanwhile, ISIS LSP flooding can be caused by issues with the ISIS protocol configuration or hardware-related issues. The occurrence of these failure domains can be exacerbated by the Arista Access Node's role as a critical component of the network's edge infrastructure, making it essential to promptly address and resolve any issues that arise.

Pre-requisite Checks

- Verify that the Whitefield POP site is accessible and that all necessary personnel and equipment are on site.

- Confirm that all redundant links are operational and that there are no issues with the network's core infrastructure.

- Ensure that all necessary permissions and access rights have been granted to perform the required commands and procedures.

Potential Root Causes

Cause	Probability
Misconfigured BGP routing policy	High
Suboptimal network design	Medium
Hardware-related issue	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp, Expected Output: The output should indicate a stable BGP session with no route flapping., Description: This command is used to verify the stability of the BGP session and check for any route flapping issues.

Command: show ospf neighbor, Expected Output: The output should indicate stable OSPF adjacencies with no adjacency loss., Description: This command is used to verify the stability of the OSPF adjacencies and check for any adjacency loss issues.

Command: show isis database, Expected Output: The output should indicate a stable ISIS database with no LSP flooding., Description: This command is used to verify the stability of the ISIS database and check for any LSP flooding issues.
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Resolution Steps

Step 1: Step: Verify and correct BGP routing policy configuration, Command: show ip bgp | grep | match , Description: This command is used to verify the BGP routing policy configuration and correct any misconfigurations that may be causing the issue.

Step 2: Step: Verify and correct OSPF adjacency configuration, Command: show ospf neighbor | grep , Description: This command is used to verify the OSPF adjacency configuration and correct any misconfigurations that may be causing the issue.

Step 3: Step: Verify and correct ISIS LSP flooding configuration, Command: show isis database | grep , Description: This command is used to verify the ISIS LSP flooding configuration and correct any misconfigurations that may be causing the issue.

Post-Check Verification

```
Command: show ip bgp | grep | match , Expected Output: The output should indicate a stable BGP session with no route flapping., Description: This command is used to verify the stability of the BGP session and check for any route flapping issues.

Command: show ospf neighbor | grep , Expected Output: The output should indicate stable OSPF adjacencies with no adjacency loss., Description: This command is used to verify the stability of the OSPF adjacencies and check for any adjacency loss issues.

Command: show isis database | grep , Expected Output: The output should indicate a stable ISIS database with no LSP flooding., Description: This command is used to verify the stability of the ISIS database and check for any LSP flooding issues.
```

Roll-back Procedures

- Step: Restore BGP routing policy configuration to previous state, Command: copy running-config startup-config, Description: This command is used to restore the BGP routing policy configuration to its previous state in case the fix fails.

- Step: Restore OSPF adjacency configuration to previous state, Command: copy running-config startup-config, Description: This command is used to restore the OSPF adjacency configuration to its previous state in case the fix fails.

- Step: Restore ISIS LSP flooding configuration to previous state, Command: copy running-config startup-config, Description: This command is used to restore the ISIS LSP flooding configuration to its previous state in case the fix fails.